

2025 AMC 10B Problem 23 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 23



Problem:

A rectangular grid of squares has 141 rows and 91 columns. Each square has room for two numbers. Horace and Vera each fill in the grid by putting the numbers from 1 through $141 \times 91 = 12,831$ into the squares. Horace fills the grid horizontally: he puts 1 through 91 left to right in row 1, 92 through 182 in row 2, and so on. Vera fills the grid vertically: she puts 1 through 141 top to bottom into column 1, then 142 through 282 into column 2, and so on. How many squares get two copies of the same number?

Answer Choices:

- A. 7
- B. 10
- C. 11
- D. 12
- E. 19

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Solution:

Let (x, y) represent the column and row at which the two numbers are equal.

$$H = 91(y - 1) + x$$

$$V = 141(x - 1) + y$$

Equating, we get $91(y - 1) + x = 141(x - 1) + y$, or

$$91y - 91 + x = 141x - 141 + y \implies 140x - 90y = 50 \implies 14x - 9y = 5.$$

Working mod 9, we get

$$14x - 9y \equiv 5 \pmod{9} \implies 5x \equiv 5 \pmod{9} \implies x \equiv 1 \pmod{9}.$$

Let $x = 9k + 1$. Then,

$$14(9k + 1) - 9y = 5 \implies 9y = 14 \cdot 9k + 14 - 5 = 126k + 9 \implies y = 14k + 1.$$

We require

$$1 \leq x \leq 91 \implies 1 \leq 9k + 1 \leq 91 \implies 0 \leq k \leq 10$$

$$1 \leq y \leq 141 \implies 1 \leq 14k + 1 \leq 141 \implies 0 \leq k \leq 10.$$

Thus there are a total of 11 values of k , implying there are (C) 11 cells where both of them write the same number.

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